

COMPUTER NETWORKS

MODULE 1

Content

Introduction to Computer Networks

- ❑ Components of a computer network:
Classification of networks, Network topologies, networking devices (Hub, Switch, Routers, Firewall, Gateway, NIC, Repeater).
- ❑ Switching Techniques, Multiplexing, OSI Reference Model, Introduction to TCP/IP Protocol Suite, Comparison between OSI & TCP/IP Protocol Suite.

Recommended Books

1. A.S. Tanenbaum, “Computer Networks”, Pearson Education, 5th Edition.
2. B.A. Forouzan, “Data Communications and Networking”, 5th edition, McGraw Hill
3. James F. Kurose, Keith W. Ross, “Computer Networking, A Top-Down Approach Featuring the Internet”, 6th edition, Addison Wesley
4. B.A. Forouzan, “TCP/IP Protocol Suite”, 4th edition, McGraw Hill

MODULE 1

Components of a computer network & Classification of networks

LEARNING OBJECTIVES

Define what a computer network is.

Discuss the advantages and disadvantages of computer networks.

Describe the main types of computer network.

Imagine today at 9:00 AM the Internet completely stopped working worldwide.

Can you attend online classes?

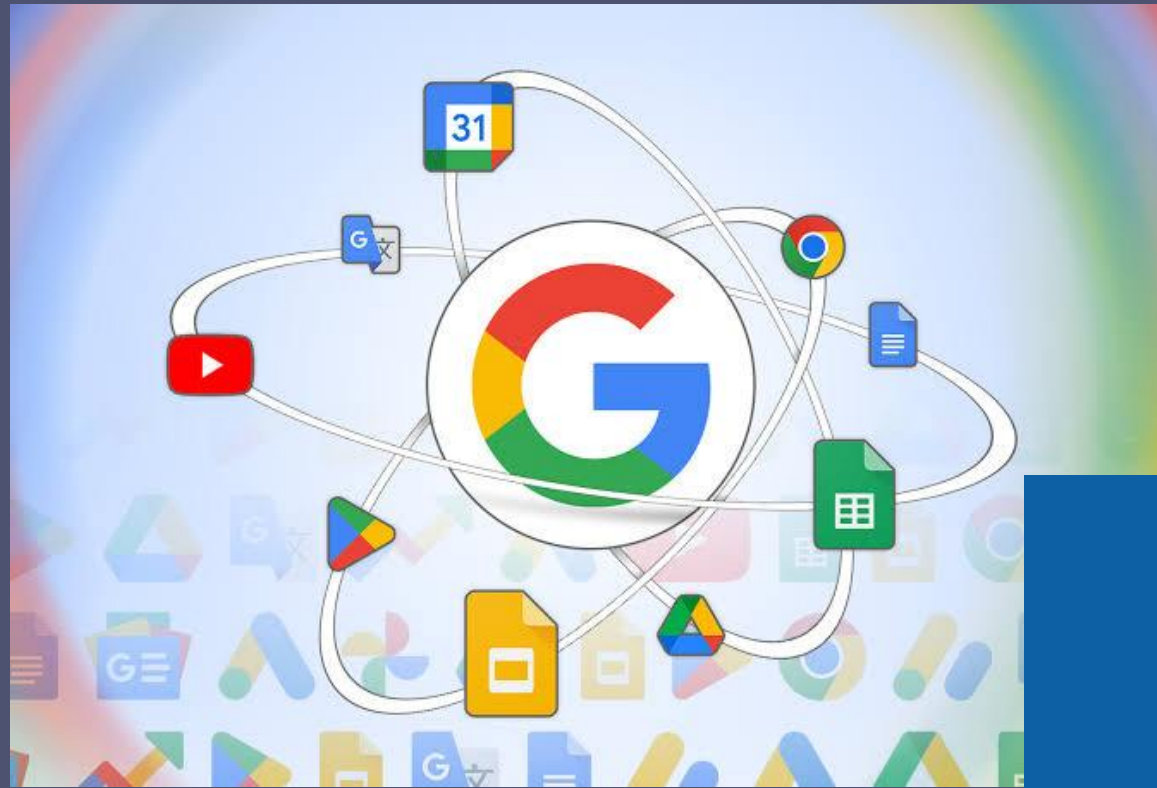
Can you use Google?

Can UPI payments work?

Can hospitals access patient records?

Every one of these services depends on computer networks.

They are all network companies. Without networking, none of these services could reach users.

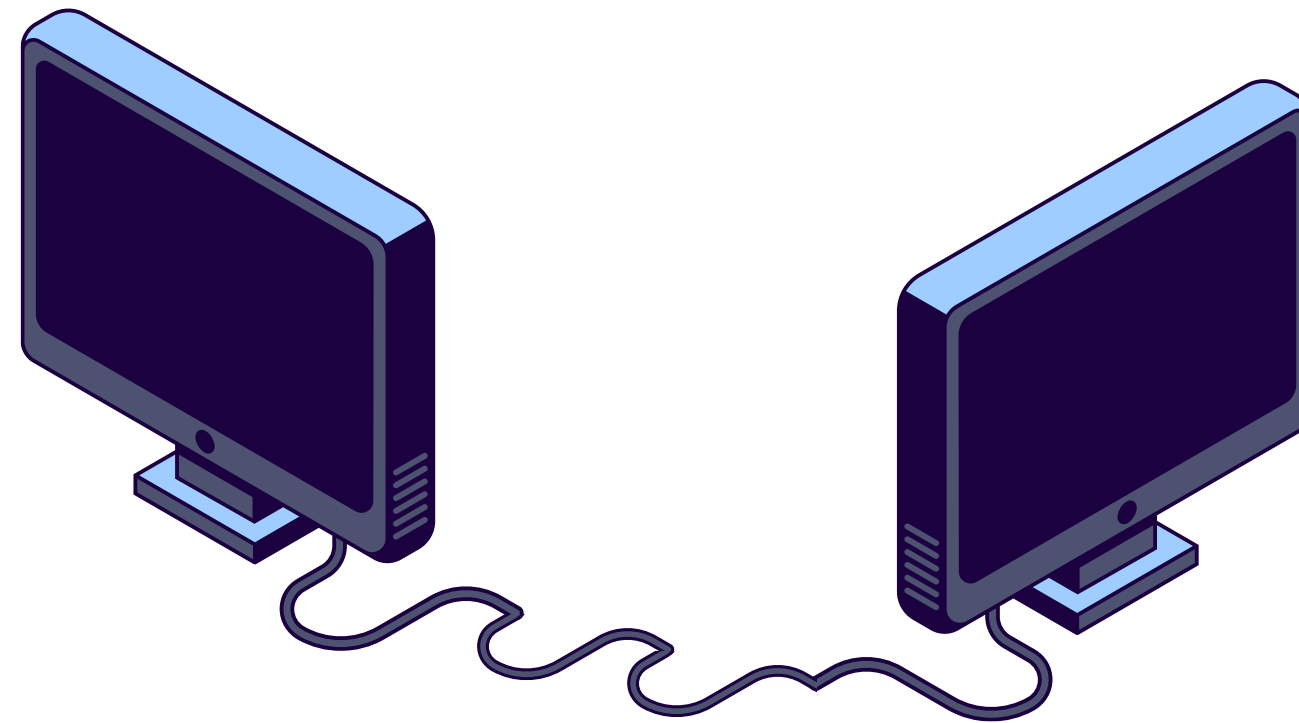


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COMPUTER NETWORKS

A computer network is two or more computers connected together, whereas a computer that is not connected to a network is stand-alone.



The term 'computer' can mean a range of different devices, including laptops, tablets and smartphones.

COMPUTER NETWORKS

- A computer network is an *interconnected* collection of *autonomous* computers.
 - ❖ Two computers are *interconnected* if they are able to *exchange* information
 - ❖ Two computers are *autonomous* if they are capable of *operating independently*, that is, neither is capable of forcibly starting, stopping, or controlling the other
- *Computer n/ws are created by different entities. Standards are needed so that these heterogeneous networks can communicate with one another.*

COMPUTER NETWORK COMPONENTS



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❖ Hardware

- ❖ “physically” connects machines (can send signals)
- ❖ eg. NIC, MEDIA, Intermediate Devices etc..

• Software

- Protocols specify services the network uses.
- Make the network hardware convenient
- eg. NOS, TCP/IP etc.

Why are Networks used by People?

❖ Resource Sharing

- ❖ printers, terminals, special architectures

❖ Information Sharing

- ❖ e-mail, world wide web

❖ Improve Reliability

❖ Improve Power (per cost)

- ❖ Networked PC's as powerful as a mainframe
- ❖ A Powerful communication medium.

Why are Networks used by People?

• Why are Networks used by People? •

Networks connect people, devices and information to share, communicate and collaborate anytime, anywhere.

1. Communication

Networks allow people to communicate instantly through emails, chats, video calls and social media.



7. Global Connection

Networks bring people and communities together across the world, breaking geographical barriers.



2. Access to Information

People can access vast amounts of information on the internet for learning, research, news and updates.



6. Convenience in Daily Life

From online banking and shopping to booking travel and paying bills, networks make everyday tasks easier and faster.



3. Education & Learning

Networks enable online learning, virtual classrooms, digital libraries and collaboration on projects.



4. Business & Work

Networks support remote work, file sharing, cloud services, VoIP calls and collaboration tools, boosting productivity.



5. Entertainment

People enjoy streaming movies, music, online gaming and connecting with others on social platforms.



In short: Networks make our lives connected, informed, productive and entertained. They power the digital world we live in today.



Effect on Society

Information Superhighway

- High-speed Internet
- Cloud Computing
- IoT & Smart Cities

Electronic “conversations

- Email
- Instant Messaging
- Video Conferencing
- Social Media

World Wide Web

- ✓ instant sharing of information
- ✓ E-Commerce
- ✓ E-Learning
- ✓ Digital Banking

Business & Education

- Remote Work
- MOOCs
- Digital Libraries
- Cloud Collaboration

Challenges

- Cybersecurity
- Privacy Issues
- Fake News
- Data Breaches

Networks connect people, businesses, governments, and devices, enabling the digital economy while creating new security and privacy challenges.



EFFECT ON SOCIETY



1. INFORMATION SUPERHIGHWAY



- High-speed networks connecting people, organizations and systems
- Global access to vast amount of information
- Enables cloud computing and remote collaboration
- Supports IoT, smart cities and Industry 4.0
- Foundation for 5G and AI-driven applications

2. ELECTRONIC "CONVERSATIONS"



- Email
- Instant Messaging (WhatsApp, Slack, Microsoft Teams)
- Video Conferencing (Zoom, Google Meet)
- Social Media Communication (Facebook, Instagram, X)
- Supports real-time and asynchronous communication
- Enables global collaboration and remote work

3. WORLD WIDE WEB



- Instant sharing of information
- True "desk-top-publishing"
- Electronic retailing (E-Commerce)
- E-learning (Coursera, NPTEL, Moodle)
- Digital Banking & UPI
- Cloud Services (Google Drive, OneDrive)
- Online Healthcare (Telemedicine)
- Streaming Services (YouTube, Netflix)

4. IMPACT ON BUSINESS & INDUSTRY



E-Commerce
(Amazon, Flipkart)

Digital
Marketing

Online
Payments
(UPI, Wallets)

Global Supply
Chain
Management

Work From
Home
(WFH)

5. IMPACT ON EDUCATION



Virtual
Classrooms

Online
Examinations

Digital
Libraries

MOOCs
(NPTEL, SWAYAM,
Coursera)

AI-based
Learning
Platforms

6. CHALLENGES & CONCERNS



Cybersecurity Threats

Digital Addiction

Privacy Concerns

Identity Theft

Fake News &
Misinformation

Data Breaches

7. REAL-WORLD APPLICATIONS



Banking
Internet Banking,
UPI, Mobile
Banking

Healthcare
Telemedicine,
Online
Consultation

Education
Online Learning,
Virtual Labs,
E-Resources

Business
ERP, CRM,
Cloud
Computing

Government
Digital India,
E-Governance,
Online Services

Entertainment
Netflix, YouTube,
Gaming, Music
Streaming

Transportation
GPS Navigation,
Ride Sharing,
Traffic Updates

KEY TAKEAWAY



Computer Networks have transformed communication, education, healthcare, business, entertainment, and governance by enabling fast, reliable, and secure information exchange across the globe.



The Information Superhighway connects the world, empowers innovation, drives the digital economy, and brings immense opportunities while posing new challenges.



ADVANTAGES OF NETWORKING

Files are stored centrally on a server so can be accessed on computers across the network.



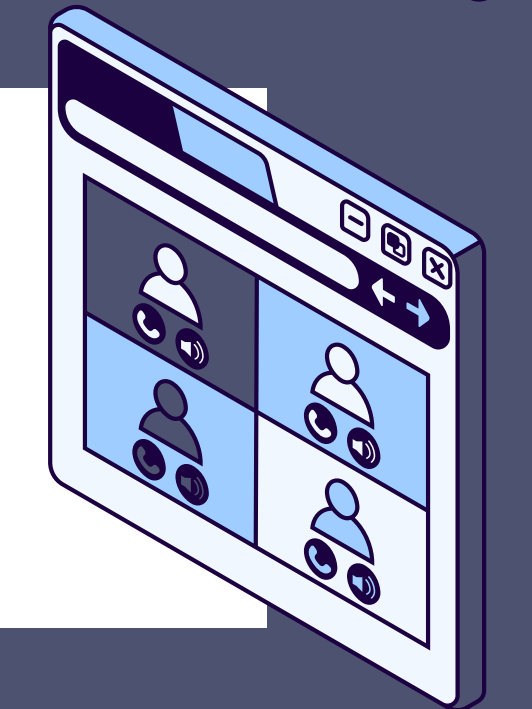
One internet connection can be used for multiple different devices using a router.



Devices (e.g. printers, scanners) are accessible on any network connected computer.

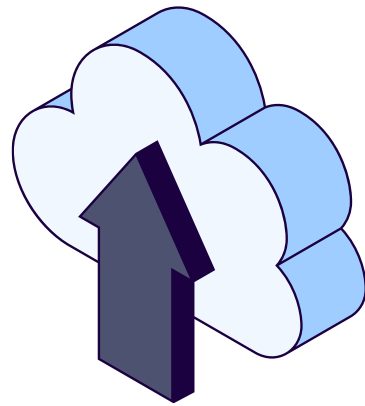


User accounts and software can be deployed across network computers.



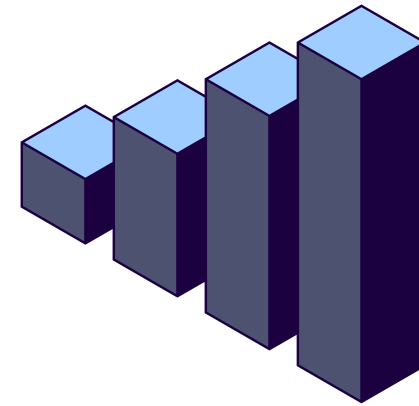
DISADVANTAGES OF NETWORKING

Availability



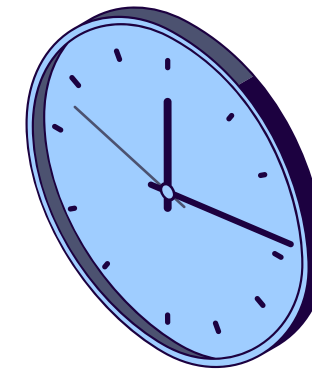
If connection is lost,
files and devices are
unavailable.

Performance



Larger networks are
slower and less
reliable.

Management



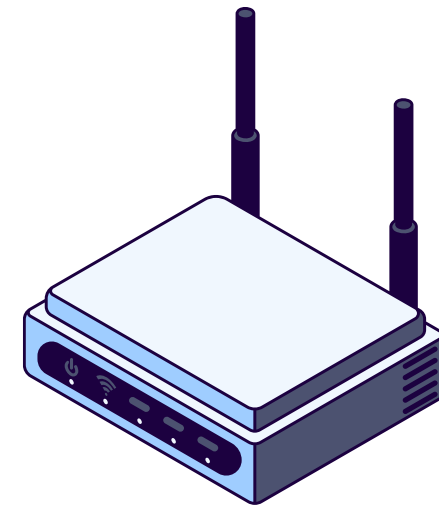
Specialist knowledge
and skills are
required.

PAN



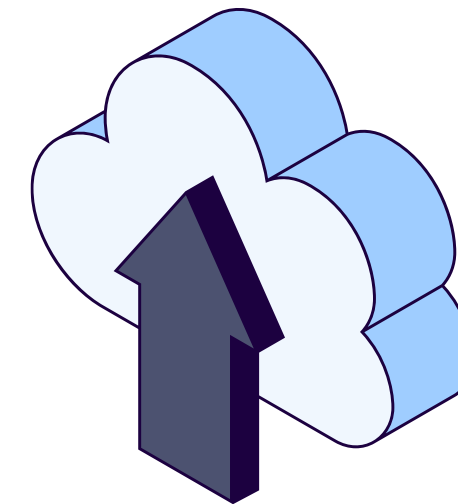
Personal Area
Network

LAN



Local Area
Network

WAN



Wide Area
Network

TYPES OF NETWORKS

PERSONAL AREA NETWORK

PAN (Personal Area Network) is used within a very small area. This is usually on a person or in a vehicle.

Bluetooth is usually used to connect these devices, which has a typical range of about 10 metres.

Examples of PANs:

- smartwatches
- wireless headphones
- wireless keyboards/mice
- car entertainment systems

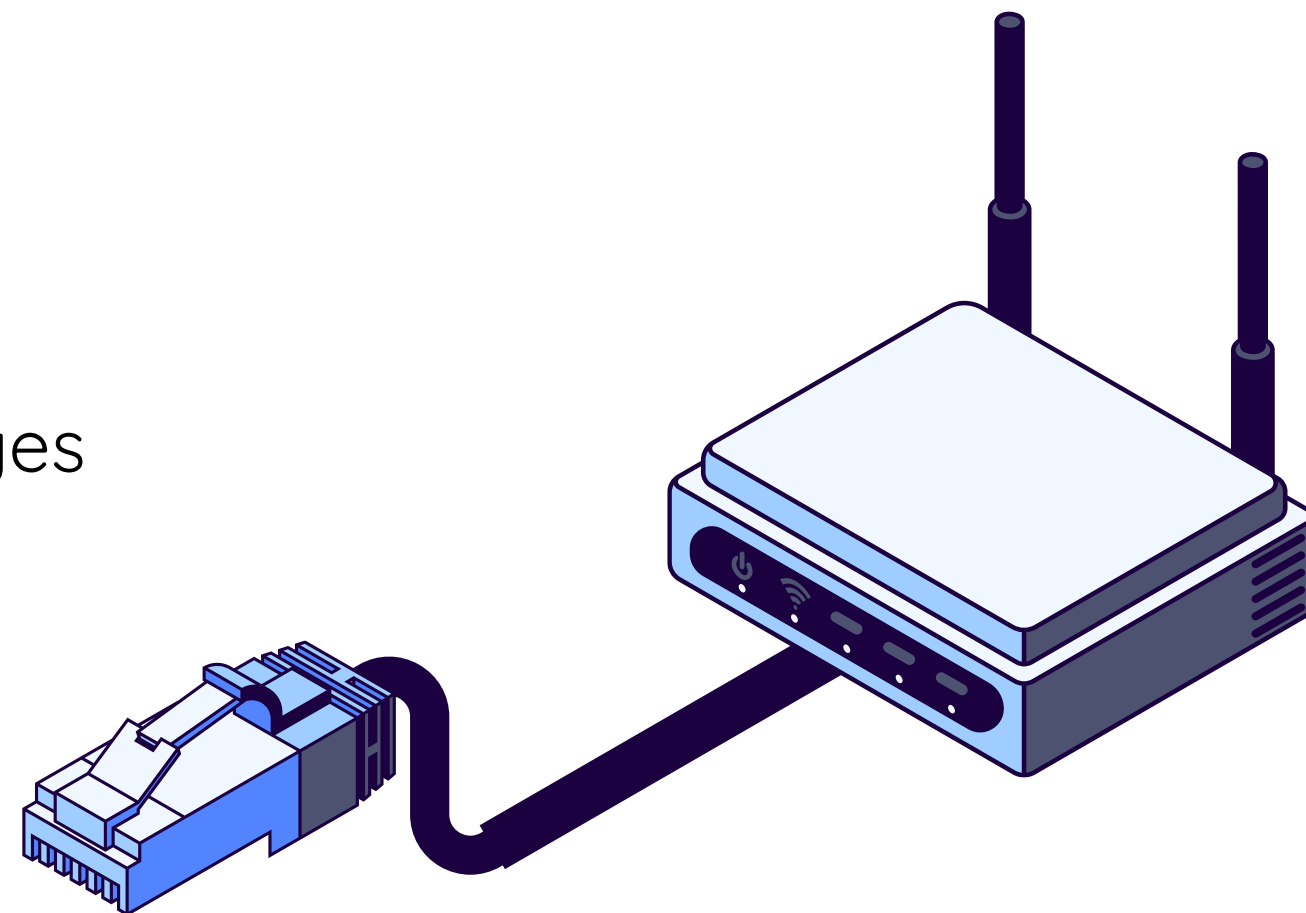


LAN (Local Area Network) is used within a small geographical area, which is usually limited to one building.

These networks can be connected wirelessly but for reliability, wired hardware (Ethernet) is often used.

Examples of LANs:

- home networks
- small businesses
- schools and colleges



LOCAL AREA NETWORK

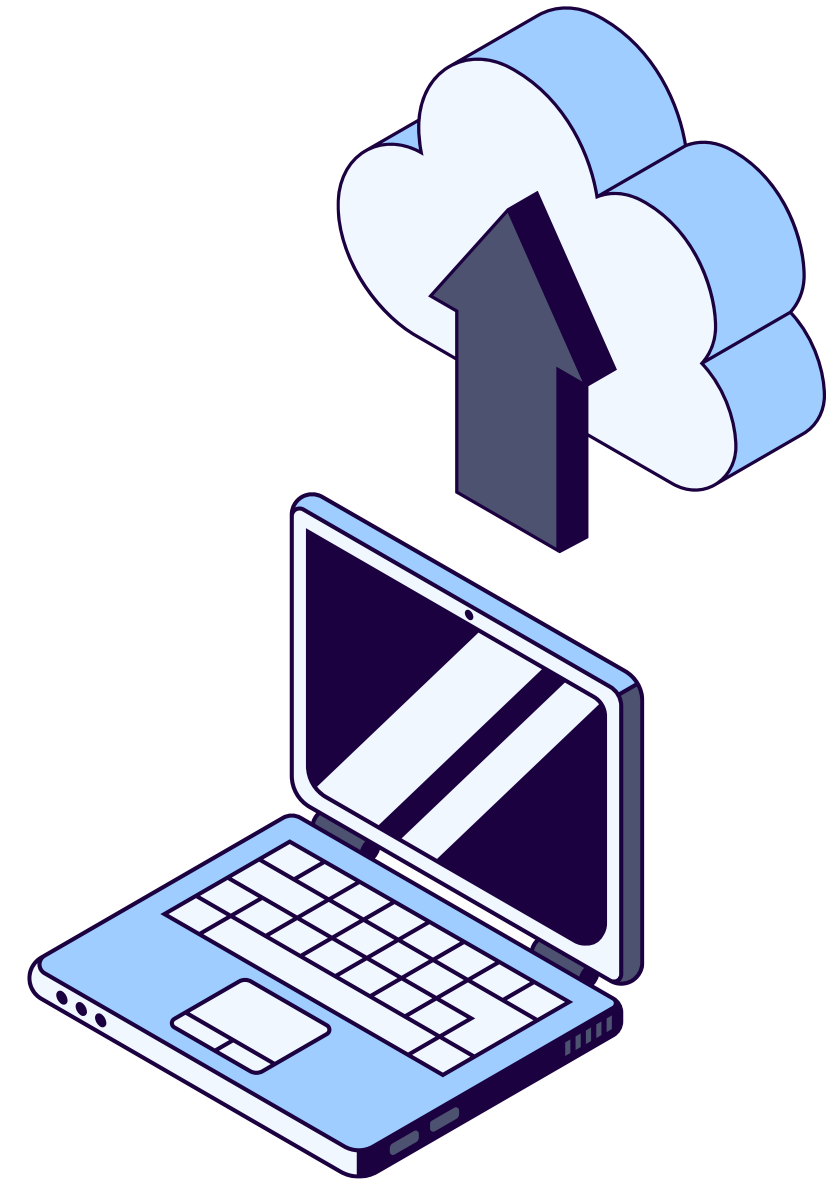
WIDE AREA NETWORK

WAN (Wide Area Network) is spread across a large geographical area, often an entire country or further.

The internet is a large scale WAN, which uses undersea cables to connect computers across the world.

Examples of WANs:

- the internet
- mobile networks
- banking networks



METROPOLITAN AREA NETWORK

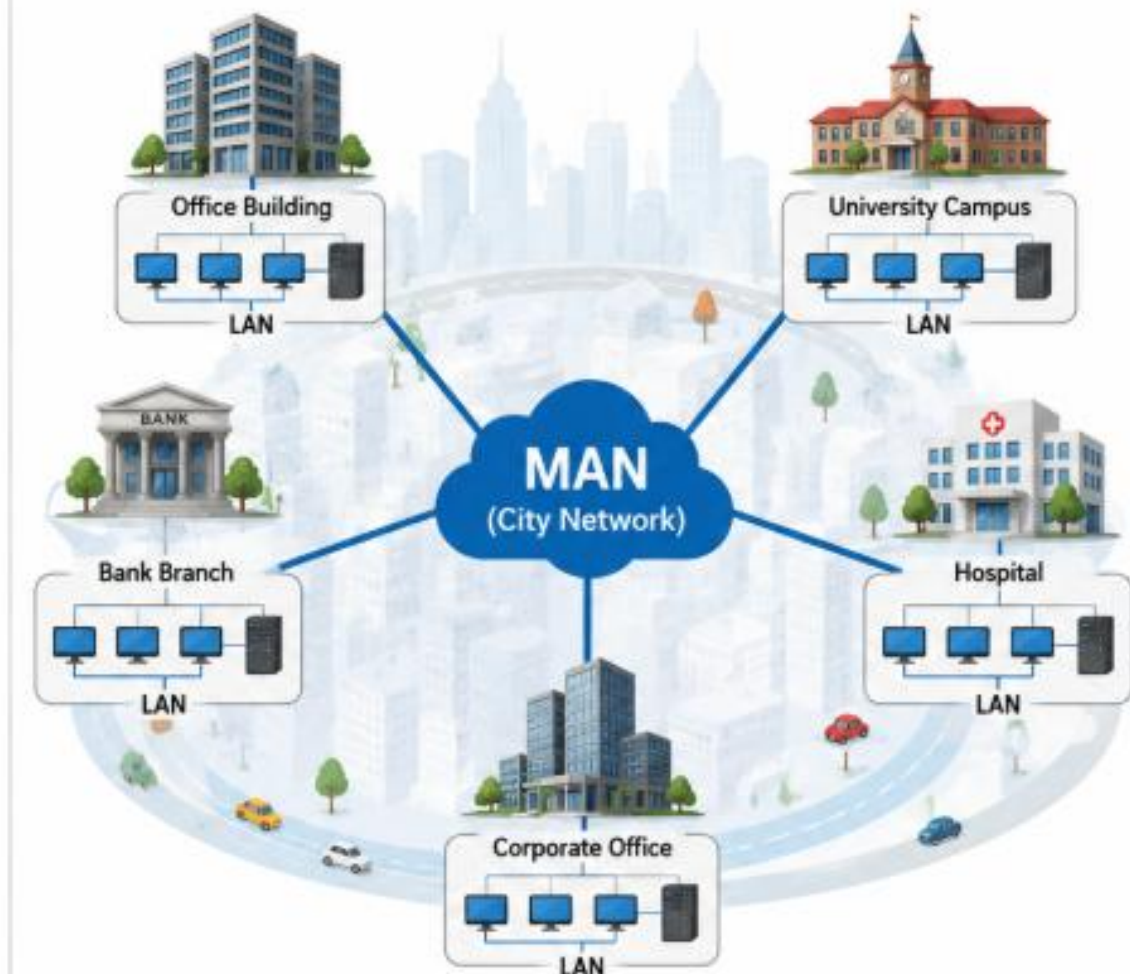
MAN (Metropolitan Area Network) is a network that covers a large metropolitan area such as a city or a large campus.

It is larger than a LAN but smaller than a WAN. MAN connects multiple LANs across a city using high-speed links (fiber optic, leased lines, microwave, etc.).

Examples of MANs:

-  City-wide government networks
-  University/college campus network connecting multiple buildings
-  Bank networks connecting different branches in a city
-  Hospital networks connecting different hospitals/centers
-  Enterprise networks connecting multiple offices in a city

MAN – Connects Multiple LANs Across a City



 Coverage: Covers a city or metropolitan area (10 – 50 km)	 Speed: High-speed Connections (100 Mbps to multiple Gbps)	 Used By: Organizations that operate across a city	 Purpose: Connect multiple LANs for sharing resources and information
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Key Point: MAN acts as a backbone network that interconnects multiple LANs within a city, enabling high-speed communication and centralized services.

METROPOLITAN AREA NETWORK

- For a town or city
- Consists of multiple LANs
- larger than local-area networks (LANs)
- but smaller than wide-area networks (WANs)
- characterized by very high-speed connections using fiber optical cable or other digital media.
- **Examples:** *part of telephone company n/w that can provide a high speed DSL line to the customer.*
 - *Cable TV n/w that originally was designed for cable TV, but today can also be used for high-speed data connection to the Internet.*



Drag and drop the network examples into the correct category.

Smartwatch	Office network	Bank network
Home network	In-car systems	Headphones
School network	The internet	Mobile network

PAN

LAN

WAN

Drag and drop the
network examples into
the correct category.

PAN

Smartwatch

Headphones

In-car systems

LAN

Home network

Office network

School network

WAN

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Bank network